

AYUSH KUMAR BHADANI

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[LinkedIn Profile](#) [GitHub](#) [LeetCode](#)

Education

Bangalore Institute of Technology

2023-2027

Bachelor of Technology in Computer Science and Engineering (Data Science) | CGPA: 9.36 Bangalore, Karnataka

Delhi Public School

2022

Senior Secondary Education in Science (PCM) | Percent: 87.2 Ranchi, Jharkhand

St. Michael's School

2020

Secondary Education | Percent: 95.6 Ranchi, Jharkhand

Relevant Coursework

- Data Structures and Algorithms
- Object-Oriented Programming
- Computer Networks
- Database Management Systems
- Operating Systems
- Machine Learning

Projects

CodeVerdict | Python, FastAPI, Groq/Llama 3, FAISS, RAG, Agentic AI

Live App

Designed, built, and orchestrated a **multi-agent AI workflow** with **4 specialized async agents** (Security, Quality, Performance, Test Coverage) using Python **asyncio** as the **agentic AI framework**, enabling concurrent execution with isolated error handling per agent.

Developed a **Retrieval-Augmented Generation (RAG)** system using **Sentence Transformer** embeddings, **FAISS** vector search, and **prompt engineering** to ground LLM agent outputs in secure-coding reference documentation, processing **unstructured text** (code files, PRs, documentation).

Integrated **Bandit static analysis (SAST)** into the Security Agent's **LLM** pipeline with a deterministic merge-verification step, ensuring static analysis findings are never omitted from the final synthesized output.

Designed and ran a **precision/recall evaluation suite** against a labeled PR benchmark dataset, achieving **87.5% precision** and **100% recall**, with documented failure-mode analysis and a tracked mitigation plan.

Digital Wellbeing Intelligence System | Python, scikit-learn, Plotly, FastAPI, NLP

Live App

Architected an end-to-end **ML pipeline** covering **data preparation, feature engineering, model training, and evaluation**, applying **K-Means clustering** to segment users into **3 behavioral risk profiles** with **~85% consistency**, enabling targeted wellness interventions.

Engineered **4 behavioral metrics** from structured and **unstructured user data** applying feature normalization to reduce intra-cluster variance by **~30%** and improve segmentation stability across test runs.

Processed and cleaned **10K+ behavioral records** using **Pandas** and **NumPy**; developed automated **data visualization** dashboards in Plotly (gauge charts, radar plots), reducing manual reporting effort by **~40%**.

Developed scalable **REST APIs** and automation scripts using **FastAPI** with asynchronous request handling (**~200ms** inference latency), streamlining the data pipeline and enabling enterprise-scale deployment.

Technical Skills

Programming Languages: Python, C++, C, SQL

AI/ML & NLP: scikit-learn, TF-IDF, Sentence Transformers, K-Means, Logistic Regression

Databases & Tools: MongoDB, MySQL, Git, GitHub, Streamlit, VS Code

Data & Visualization: Pandas, NumPy, Matplotlib, Plotly, Power BI, Tableau

Methodologies: SDLC, Agile/Scrum, EDA

Soft Skills: Analytical Thinking, Problem Solving, Technical Communication, Leadership

Achievements & Activities

VTU Branch Topper – 3rd Semester | CGPA: 9.86

Secured the highest academic performance in the **CSE (Data Science)** branch at Bangalore Institute of Technology.

Core Technical Member | Google Developer Group (GDG)

Collaborated with a **5-member team** to deliver the official GDG website across sprints; conducted **3+ workshops** and **2+ regional events** reaching **1000+ participants**.

IoT Exhibition – Consolation Prize | Arduino, Embedded Systems, Automation

Built an **automated solar panel cleaning system** using **Arduino** with motor control and dust-sensor feedback, demonstrating applied **IoT and embedded automation** skills.

Leadership & Outreach

Organized **amBITion 2.0** and **AI Study Jams** for **200+ participants**; represented college at **inter-college hackathons** and mentored **10+ junior students** on DSA and Python fundamentals.